



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Load Balancer: Intelligent load balancing and resource allocation

Team ID: T140

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: T140

Semester: 3rd

Section: J,DS-2,DS-1,Cyber

Domain: *DS+CPP*

Team Members

1. *Amrit Raj – 2510010756*
2. *Pawani Gupta – 2510380107*
3. *Anvi Shandilya – 2510380201*
4. *Rushil Kumar – 2510350004*

Mentor

Mr.Piyush Agarwal

Interactions completed: _____

Problem Statement & Motivation

Problem Statement

Servers have to handle a lot of task at a time . Uneven workload distribution can overload some servers while leaving others underutilized, resulting in higher response time and inefficient resource usage. A loadbalancer aims to distribute incoming tasks among available resources based on their current workload.

Motivation

- ▶ *Uneven workload distribution among servers.*
- ▶ *Overloaded servers can reduce system performance.*
- ▶ *Simple methods may not handle*

Target Users / Stakeholders

- ▶ *Administrators*
- ▶ *Server users*
- ▶ *Better resource utilizations*

Objectives

Project Objectives

1. *Develop a system to balance workloads along different servers.*
2. *Allocate task based on server load and resource availability.*
3. *Integrate DSA in C and OOP with C++ concepts.*
4. *Test and evaluate the system using suitable test case.*
5. *Reduce server load.*

Key Objective: *The main measurable outcomes of a load balancer focus on performance, reliability, efficiency, and security.*

Proposed Solution

Proposed Approach

1. *Collect server details, traffic on them*
2. *Analyze server load and select best server*
3. *Assign tasks according to server availability and load on them .*
4. *Stores server allocation details*
5. *Display server load details*

Key Features

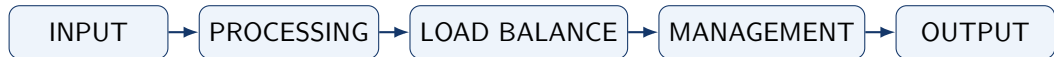
- ▶ *Load based task allocation*
- ▶ *Efficient resource utilization*
- ▶ *Server load monitoring*
- ▶ *Dynamic task distribution*
- ▶ *Simple result visualization*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Arrays, Queue, Priority Queue, Searching and Sorting, Structure.	DSA with C
OOPs with C++	Classes and Objects, Encapsulation, Inheritance, Abstraction, File Handling.	OOP using C++

Requirement: The project should integrate concepts from both PBL subjects and show how they work together in making of Load Balancer , from detecting traffic do distributing it **among** servers.

System Architecture / Workflow



Major Components

- ▶ *Server Task Input Module*
- ▶ *Load Balancing Module*
- ▶ *Resource Module*

Data / Control Flow

- ▶ *User provides task and server details*
- ▶ *System Analyse Server Workload*
- ▶ *Task is allocated and result is displayed*

Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C / C++*
- ▶ Database: *Not Required*
- ▶ Tools / Framework: *Std C/C++ Libraries*
- ▶ IDE: *Visual Studio Code*
- ▶ Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

Core concepts of DSA with C and OOP with C++. VS - CODE and GITHUB makes development and collaboration easier

Initial Progress & Team Contribution

Progress Achieved

- ▶ *Problem studied*
- ▶ *Requirements identified*
- ▶ *Architecture prepared*
- ▶ *Initial module implemented*
- ▶ *Solution finalised*

Individual Contribution

Member	Role	Contribution
Amrit Raj	Lead	C - Dsa developer/Research
Pawani Gupta	Developer	Oops Developer
Anvi	DB/Testing	C-dsa developer/Testing
Shandilya		
Rushil Kumar	Developer/Docs	Oops Developer/Presentation

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ *Working prototype / system*
- ▶ *Measurable results*
- ▶ *Practical applicability*
- ▶ *Future scalability*

References

1. *Cloud Load Balancing - IEEE Xplore*
2. *Cloud Computing - Thomas erl*
3. *AWS Elastic Load Balancing*
4. *GitHub - Cloud Load Balancer*

Thank You

Questions & Suggestions